

# Apêndice B – Protocolo de comunicação Externo (Interface)

## A. Communication Protocol

Information is transferred by the following methods.

<SB>information<EB><CR>

<SB> is Start Block Character      needs 1byte    corresponds to ASCII <VT>  
hexadecimal 0x0B

<EB> is End Block Character      needs 1byte    corresponds to ASCII <FS>  
Hexadecimal 0x1C

<CR> is Carriage Return      needs 1byte    corresponds to ASCII <CR>  
hexadecimal 0x0D

Information is the data that we want to transfer. Please refer to the following for details.

## B. Information Grammar

### 1. Delimiter

| ---    Fields Delimiter  
^ ---    Component Delimiter  
& ---    Subcomponent Delimiter  
~ ---    Repeat Delimiter  
\ ---    Escape Character

### 2. Data Type

CX extended composite id which check digit  
CE code element  
CM composite  
CQ composite quantity with units  
DR date time range  
DT data  
DLN driver's license number  
EI entity identifier  
HD hierarchic designator  
FN family name  
FT formatter text  
IS coded value for user-defined tables  
ID coded values for HL7 tables  
JCC job code  
NM numeric  
PT processing type  
PL person location  
ST string  
SI sequence ID  
TS time stamp  
TQ timing quantity  
TX text data  
XAD extended address  
XCN extended composite ID number and name  
XON extended composite name and ID number for organizations  
XPN extended person name  
XTN extended telecommunications number  
VID version identifier

### 3. Field Meaning

#### 3.1. There is a message header at the beginning of each message. It is MSH field.

The meaning of MSH is shown as below

| No. | Field                           | Data Type | Length | Explanation                                                                        |
|-----|---------------------------------|-----------|--------|------------------------------------------------------------------------------------|
| 1   | Field mark                      | ST        | 1      | Separator                                                                          |
| 2   | Encoding chars                  | ST        | 4      | Separator listing                                                                  |
| 3   | Sending Application             | EI        | 180    | Sending end applications                                                           |
| 4   | Sending Facility                | EI        | 180    | Sending end facility                                                               |
| 5   | Receiving Application           | EI        | 180    | Receiving end applications                                                         |
| 6   | Receiving Facility              | EI        | 180    | Receiving end facility                                                             |
| 7   | Date Time Message               | TS        | 26     | Current message event, system time                                                 |
| 8   | Security                        | ST        | 40     | Security                                                                           |
| 9   | Message Type                    | CM        | 7      | Message Type                                                                       |
| 10  | Message Control ID              | ST        | 20     | Message control ID is used to distinguish different messages. See the table below. |
| 11  | Processing ID                   | PT        | 3      | Dispose of ID P Product                                                            |
| 12  | Version ID                      | VID       | 60     | HL7 version is 2.3.1                                                               |
| 13  | Application Acknowledgment Type | IS        | 1      | Set null                                                                           |
| 14  |                                 |           |        | Retain                                                                             |
| 15  |                                 |           |        | Retain                                                                             |
| 16  |                                 |           |        | Retain                                                                             |
| 17  |                                 |           |        | Retain                                                                             |
| 18  | Encoder                         | ST        |        | Encoding is UNICODE                                                                |

| MSH-10 | Description                                              |
|--------|----------------------------------------------------------|
| 0001   | Analyzer transmits results automatically.                |
| 1001   | LIS responses, analyzer transmits results automatically. |

Example:

5160|LIS|PC|20100930100436||ORU^R01|0001|P|2.3.1|1||||UNICODE

MSH|^~\&|URIT|UT-

#### 3.2. PID--- Definition of patients' data field

| No. | Field                   | Data Type | Length | Explanation                                       |
|-----|-------------------------|-----------|--------|---------------------------------------------------|
| 1   | Set ID PID              | SI        | 4      | Identify different fields, fill with 1 generally. |
| 2   | Patient ID              | EI        | 20     | Patient ID., hospital No., set null               |
| 3   | Patient Identifier List | CX        | 20     | Indicate batch number when QC                     |
| 4   | Alternate Patient ID    | CX        | 20     | Bed No.                                           |
| 5   | Patient Name            | XPN       | 48     | Name                                              |
| 6   | Mother's Maiden Name    | XPN       | 48     | Mother's Maiden Name, set null                    |
| 7   | Date/Time of Birth      | TS        | 26     | Birthday; Indicate validity when QC               |

|     |                                           |     |     |                         |
|-----|-------------------------------------------|-----|-----|-------------------------|
| 8   | Sex                                       | IS  | 1   | Male or female          |
| 9   | Patient Alias                             | XPN | 48  | Retain patient alias    |
| 10  | Race                                      | CE  | 80  | Retain race             |
| 11  | Patient Address                           | XAD | 106 | Retain patient address  |
| 12  | County Code                               | IS  | 4   | Retain county code      |
| 13  | Phone Number                              | XTN | 40  | Retain phone No.        |
| 13  | Phone Number Bus                          | XTN | 40  | Retain office phone No. |
| 14  | Primary Language                          | CE  | 60  | Retain mother tongue    |
| 15  | Marital Status                            | CE  | 80  | Retain Marital Status   |
| 16  | Religion                                  | CE  | 80  | Retain religion         |
| ... | The rest part is not needed to be filled. |     |     |                         |

Example: PID|1|1010051|A1123145|15|Mary||19811011|M

### 3.3. PV1---Definition of patient visiting record field

| No. | Field                     | Data Type | Length | Explanation                                       |
|-----|---------------------------|-----------|--------|---------------------------------------------------|
| 1   | Set ID PV1                | SI        | 4      | Identify different fields, fill with 1 generally. |
| 2   | Patient Class             | IS        | 1      | Patient category                                  |
| 3   | Assigned Patient Location | PL        | 80     | Be used to indicate patient department            |

Example: PV1|1|Clinic| Surgery |

### 3.4. OBR--- Definition of Doctor's Advice

| No. | Field                       | Data Type | Length | Explanation                                                           |
|-----|-----------------------------|-----------|--------|-----------------------------------------------------------------------|
| 1   | Set ID OBR                  | SI        | 4      | Identify different fields, fill with 1 generally.                     |
| 2   | Placer Order Number         | EI        | 22     | Serial number                                                         |
| 3   | Assigned Patient Location   | EI        | 22     | Sample number                                                         |
| 4   | Universal Service ID        | CE        | 200    | Universal service ID                                                  |
| 5   | Priority                    | ID        | 2      | Priority set null                                                     |
| 6   | Requested Date Time         | TS        | 26     | Application time                                                      |
| 7   | Observation Date Time       | TS        | 26     | Inspection starting time, set null                                    |
| 8   | Observation Date Time end   | TS        | 26     | Inspection end time                                                   |
| 9   | Collection Volume           | CQ        | 20     | Specimen collection capacity, set null                                |
| 10  | Collector Identifier        | XCN       | 60     | Sender name                                                           |
| 11  | SPE Action Code             | ID        | 1      | Sample handling code, set null                                        |
| 12  | Danger Code                 | CE        | 60     | Danger code alarm                                                     |
| 13  | Relevant Clinical Info      | ST        | 200    | "Diagnosis" ^ "Remark", each length should not be more than 100 bytes |
| 14  | SPE Received Date Time      | TS        | 26     | Sample receiving time                                                 |
| 15  | SPE Source                  | CM        | 300    | Sample classification, blood, urine etc.                              |
| 16  | Ordering Provider           | XCN       | 120    | Inspector name                                                        |
| 17  | Order Callback Phone Number | XTN       | 40     | Callback phone, set null                                              |



|   |                             |    |     |                 |
|---|-----------------------------|----|-----|-----------------|
| 5 | Delayed Acknowledgment Type | ID | 1   |                 |
| 6 | Error Condition             | CE | 100 | Error condition |

MMSA-6 is used to indicate different errors, see the table below.

| MSA-1 | MSA-6 | MSA-3                      | False Description                                                                          |
|-------|-------|----------------------------|--------------------------------------------------------------------------------------------|
| AA    | 0     | Message accepted           | Receive successfully                                                                       |
| AE    | 101   | Segment sequence error     | The fields order in message is not correct, or the necessary fields are lost.              |
|       | 102   | Required field missing     | Necessary fields of a paragraph are lost.                                                  |
|       | 103   | Data type error            | Data type of fields is false. For example, digital is changed into character.              |
|       | 104   | Key not found              | Key identifier is not found                                                                |
|       | 105   | Resend                     | Resend data                                                                                |
| AR    | 201   | Unsupported message type   | Unsupported message type                                                                   |
|       | 202   | Unsupported event code     | Unsupported event code                                                                     |
|       | 203   | Unsupported processing id  | Unsupported processing ID                                                                  |
|       | 204   | Unsupported version id     | Unsupported version ID                                                                     |
|       | 205   | Unknown key identifier     | Unknown key identifier, For example, transmit an inexistent patient information.           |
|       | 206   | Duplicate key identifier   | Duplicate key identifier                                                                   |
|       | 207   | Application record locked  | Affairs in application storage level can't be carried out. For example, database is locked |
|       | 208   | Application internal error | Other errors in unknown application.                                                       |
|       | 209   | Application unready        | Application is not ready                                                                   |

### 3.7. ERR

| No. | Field                   | Data Type | Length | Explanation             |
|-----|-------------------------|-----------|--------|-------------------------|
| 1   | Error Code and Location | CM        | 80     | Code and position error |

### ERR-1

| Assembly 1 | Assembly 2           | Assembly 3         | Explanation                                      |
|------------|----------------------|--------------------|--------------------------------------------------|
| 001        | Record already exist | Test tube No.      | The test tube record has already existed.        |
| 002        | Lis Recieved Failed  | Test tube No.      | Lis receiving error, resending data is required. |
| 003        | Read REQ error       | Test tube No.      | Fail to read request form.                       |
| 004        | Read BarCode Error   | Test tube rack No. | Analyzer fails to read test tube number.         |

### 3.8. QRD

| No. | Field | Data Type | Length | Explanation |
|-----|-------|-----------|--------|-------------|
|-----|-------|-----------|--------|-------------|

|    |                             |     |    |                                                                                     |
|----|-----------------------------|-----|----|-------------------------------------------------------------------------------------|
| 1  | Query Date/Time             | TS  | 26 | Query time                                                                          |
| 2  | Query Format Code           | ID  | 1  | D (display format)                                                                  |
| 3  | Query Priority              | ID  | 1  | I (Immediate)                                                                       |
| 4  | Query ID                    | ST  | 10 | Distinguish different queries ,accumulate with query times. The initial value is 1. |
| 5  | Deferred Response Type      | ID  | 1  | Set null                                                                            |
| 6  | Deferred Response Date/Time | TS  | 26 | Set null                                                                            |
| 7  | Quantity Limited Request    | CQ  | 10 | RD (Records)                                                                        |
| 8  | Who Subject Filter          | XCN | 60 | Take as a test tube code \ sample number.                                           |
| 9  | What Subject Filter         | CE  | 60 | OTH                                                                                 |
| 10 | What Department Data Code   | CE  | 60 | Set null                                                                            |
| 11 | What Data Code Value Qual.  | CM  | 20 | Set null                                                                            |
| 12 | Query Results Level         | ID  | 1  |                                                                                     |

### 3.9. QRF

| No. | Field                            | Data Type | Length | Explanation                                                                     |
|-----|----------------------------------|-----------|--------|---------------------------------------------------------------------------------|
| 1   | Where Subject Filter             | ST        | 20     | Take UT-5160                                                                    |
| 2   | When Data Start Date/Time        | TS        | 26     | Application time                                                                |
| 3   | When Data End Date/Time          | TS        | 26     | Deadline                                                                        |
| 4   | What User Qualifier              | ST        | 60     | Set null                                                                        |
| 5   | Other QRY Subject Filter         | ST        | 60     | Set null                                                                        |
| 6   | Which Date/Time Qualifier        | ID        | 12     | RCT(Specimen receipt date/time, receipt of specimen in filling ancillary (Lab)) |
| 7   | Which Date/Time Status Qualifier | ID        | 12     | ANY(Any status)                                                                 |
| 8   | Date/Time Selection Qualifier    | ID        | 12     | ALL(All values within the range)                                                |
| 9   | When Quantity/Timing Qualifier   | TQ        | 60     | Set null                                                                        |

### 3.10. QSP

| No. | Field               | Data Type | Length | Explanation     |
|-----|---------------------|-----------|--------|-----------------|
| 1   | Set ID - DSP        | 4         | SI     |                 |
| 2   | Display Level       | SI        | 4      |                 |
| 3   | Data Line           | TX        | 300    | Content queried |
| 4   | Logical Break Point | ST        | 4      |                 |
| 5   | Result ID           | TX        | 20     |                 |

Use QSP-1 to distinguish different queried information in QSP fields.

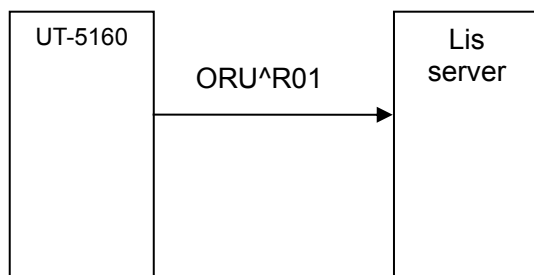
| Set ID – DSP | Message                                                 |
|--------------|---------------------------------------------------------|
| 1            | Sample SN                                               |
| 2            | Name                                                    |
| 3            | Gender                                                  |
| 4            | Age                                                     |
| 5            | Blood type                                              |
| 6            | Group                                                   |
| 7            | Patient Number                                          |
| 8            | Bed Number                                              |
| 9            | Patient Type                                            |
| 10           | Department                                              |
| 11           | Sender                                                  |
| 12           | Inspector                                               |
| 13           | Auditor                                                 |
| 14           | BLDV is for venous blood, BLDC is for peripheral blood. |
| 15           | Remark                                                  |
| 16           | Sampling time, sending time                             |
| 17           | inspection time                                         |

Example

DSP|1||Mary||<CR>

#### 4. Communication process

##### 4.1. Analyzer transmits test results to lis server



```

<SB>
MSH
PID
PV1
OBR
OBX
OBX
.....
<EB><CR>
  
```

OBX fields can be repeated. Transmitted test results include patient information, 28 parameters, 2 histograms and 2 scatter plots. The 2 histograms and 2 scatter plots are BMP format and transmitted with base64 code.

For example:

Analyzer transmits test results to lis server

```

<SB>
MSH|^~\&|URIT|UT-
5160|LIS|PC|20110627144458||ORU^R01|0001|P|2.3.1|||||UNICODE<CR>
PID|1|||||||<CR>
  
```

PV1|1|||<CR>  
OBR|1||BAR101010101|URIT^UT-5160|||01110621143134|||^^|  
OBX|1|NM|WBC||110.0|10^9/L|40.0-100.0|H|||F|||<CR>  
OBX|2|NM|LYM||35.57|%|20.00-40.00|||F|||<CR>  
OBX|3|NM|MON||5.84|%|3.00-8.00|||F|||<CR>  
OBX|4|NM|NEU||57.37|%|50.00-70.00|||F|||<CR>  
OBX|5|NM|EOS||1.14|%|0.50-5.00|||F|||<CR>  
OBX|6|NM|BASO||0.08|%|0.00-1.00|||F|||<CR>  
OBX|7|NM|LYM#||284.5|10^9/L|80.0-400.0|||F|||<CR>  
OBX|8|NM|MON#||46.7|10^9/L|10.0-80.0|||F|||<CR>  
OBX|9|NM|NEU#||458.9|10^9/L|200.0-700.0|||F|||<CR>  
OBX|10|NM|EOS#||9.1|10^9/L|0.0-50.0|||F|||<CR>  
OBX|11|NM|BASO#||0.6|10^9/L|0.0-10.0|||F|||<CR>  
OBX|12|NM|RBC||4.49|10^12/L|3.50-5.50|||F|||<CR>  
OBX|13|NM|HGB||0|g/L|0-1079738368|L|||F|||<CR>  
OBX|14|NM|HCT||26.4|%|37.0-50.0|L|||F|||<CR>  
OBX|15|NM|MCV||59.0|fL|80.0-100.0|L|||F|||<CR>  
OBX|16|NM|MCH||24.0|pg|27.0-31.0|L|||F|||<CR>  
OBX|17|NM|MCHC||0|g/L|0-1081344000|H|||F|||<CR>  
OBX|18|NM|RDW\_CV||16.1|%|11.5-14.5|H|||F|||<CR>  
OBX|19|NM|RDW\_SD||45.0|fL|35.0-56.0|||F|||<CR>  
OBX|20|NM|PLT||0|10^9/L|0-1079574528|H|||F|||<CR>  
OBX|21|NM|MPV||12.3|fL|7.0-11.0|H|||F|||<CR>  
OBX|22|NM|PDW||14.7|fL|15.0-17.0|L|||F|||<CR>  
OBX|23|NM|PCT||0.41|%|0.10-0.28|H|||F|||<CR>  
OBX|24|NM|P\_LCR||1.37|%|0.50-1.80|||F|||<CR>  
OBX|25|ED|RBCHistogram||UT5160^Image^BMP^Base64^Qk32lgMAAA.....<CR>  
OBX|26|ED|PLTHistogram||UT5160^Image^BMP^Base64^Qk32lgMAAA.....<CR>  
OBX|27|ED|S0\_S10DIFFScattergram||UT5160^Image^BMP^Base64^Qk32lgMAAA.....<CR>  
OBX|28|ED|S90\_S90DDIFFScattergram||UT5160^Image^BMP^Base64^Qk32lgMAAA.....<CR>  
<EB><CR>